Quantum Pong

How do you play Pong at the nano scale?

Find out in this interactive demonstration!

By: Simão M. João

What is it?

In regular Pong, you bounce a ball off the paddles into the opponent's side. In Quantum Pong, the **ball is replaced by an electron**, which displays quantum behaviour! Use your knowledge of quantum mechanics to gain an edge over your opponent. **What quantum effects can you see?**

How to play

Quantum Pong can be played by two or three players. Two players control the paddles and try to **bounce the electron** into the opponent's side. A third player can intervene and **place obstacles** in the electron's path. The electron's wavefunction is represented on the screen, displaying the **probability of finding the electron at each point**. As some of the electron reaches the top or bottom of the screen, it is lost forever. **The player who loses more than 50% of the electron loses**. The bar on the right keeps track of this progress.

Controls

Player 1 (bottom paddle):
`W` - move paddle up
`A` - move paddle left
`S` - move paddle down
`D` - move paddle right

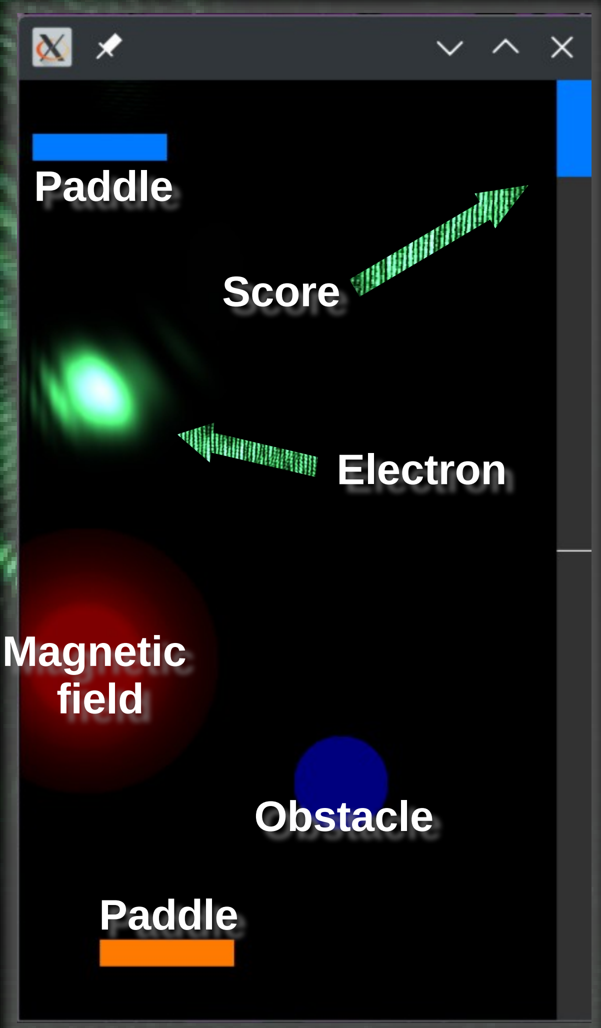
W
A S D

Player 2 (top paddle):
`I` - move paddle up
`J` - move paddle left
`K` - move paddle down
`L` - move paddle right

I
J K L

Player 3:
Mouse left button click: place obstacle
Mouse right button click: place magnetic field
`3` - increase radius of next magnetic field
`4` - decrease radius of next magnetic field

Further controls:
`R` - reset the game
`T` - freeze the scoring
`Y` - resume the scoring
`G` - freeze time
`F` - resume time
`1` - decrease the visualization threshhold
`2` - increase the visualization threshhold
Moving a paddle over an obstacle removes it



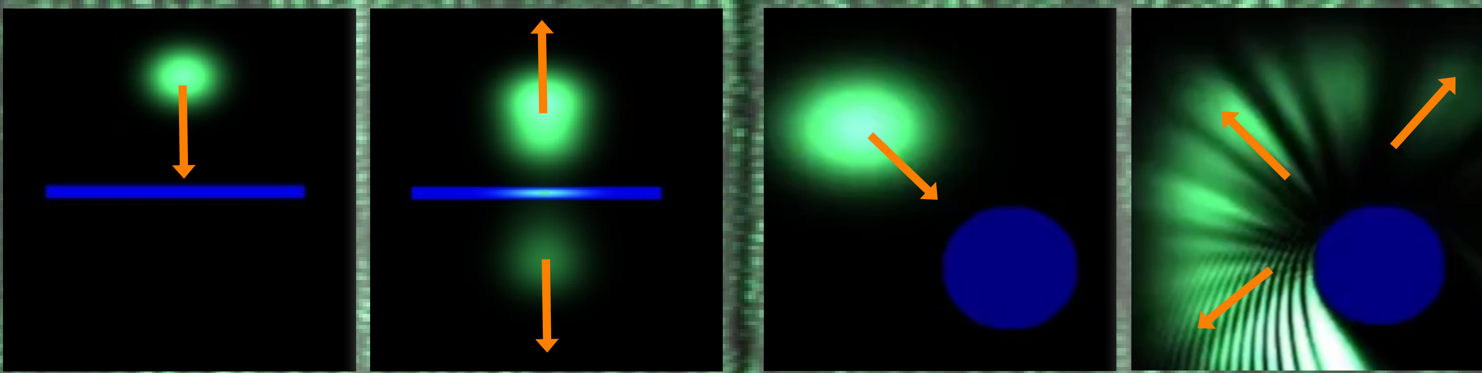
Quantum vs classical behaviour

At the nano scale, **quantum effects become the norm**. In Quantum Pong, you can interact with an electron and see it act as a **particle** and as a **wave**. If the electron is left undisturbed, it will **move forward like a particle** and **bounce off of walls like a particle**. When it interacts with objects its own size, it can display wave-like behaviour like **diffraction** and **interference**. You can alter the electron's behaviour by using certain tools, like **obstacles** and **magnetic fields**. Which of the effects you see are **quantum**, and which are **classical**?

Obstacles

If you place a small **barrier** in the electron's path, it can just **tunnel through**! You may also notice that some of the electron's wavefunction managed to get **inside the obstacle**! This is impossible in our classical world, but perfectly **possible in the quantum world**.

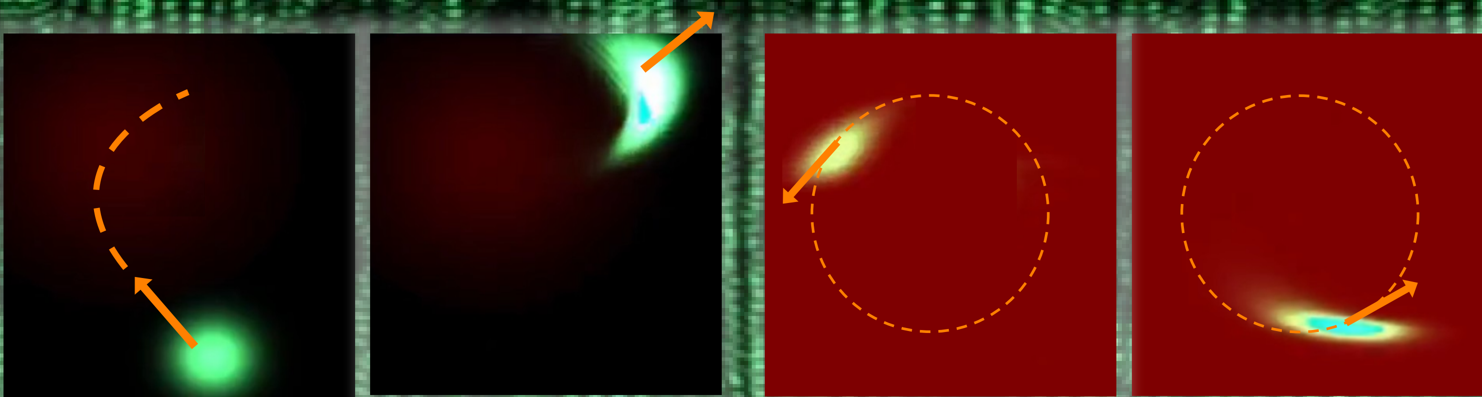
If you place an obstacle of **similar size** to the electron in its path, the bouncing turns into **scattering**, just like a wave! Now you can see **interference patterns** and **diffraction** around the obstacle.



Magnetic field

Electrons are charged particles and so they will experience the **Lorenz force** in the presence of a magnet, causing them to **deflect**.

If the magnet is **strong enough**, you can see the electron **go around in circles**. What happens if you place an obstacle now?



Want to know more?

Quantum Pong is an **open-source** game written in **C++** using **GPU acceleration** to solve **Schrödinger's equation** in real time. If you would like to get involved or are simply interested in the technical details behind how Quantum Pong works, check out the **GitHub repository** using the QR code on the right!